Modeling

Michel Bierlaire

Solution of the practice quiz

In this problem, Frank wants to maximize the number of points of his team, given some constraints. First, the decision variables are the different candidates:

x_1: Anakin,	x_{11}: Luke,
x_2: Chewbacca,	x_{12}: Obi-Wan,
x_3: Dooku,	x_{13}: Padme,
x_4: Gial,	x_{14}: Qui-Gon,
x_5: Han,	x_{15}: Sebulba,
x_6: Jabba,	x_{16}: Sheev,
x_7: Jubnuk,	x_{17}: Teebo,
x_8: Lando,	x_{18}: Watto,
x_9: Leia,	x_{19}: Yoda,
x_{10}: Lorth,	x_{20}: Zuckuss.

Let c_i denote the number of points of candidate i. Since Frank wants to maximize the number of points in his team, we get the objective function

$$\max_{x \in \{0,1\}^{20}} \sum_{i=1}^{20} c_i x_i$$

Then, we have to model each constraint:

1. The team must contain at least 3 chefs who are skilled in preparing appetizers:
$$x_5 + x_9 + x_{11} + x_{13} + x_{19} \geq 3.$$

2. The team must contain at least 4 chefs who are skilled in preparing fish:
$$x_1 + x_2 + x_6 + x_7 + x_9 + x_{12} + x_{14} + x_{19} \geq 4.$$

3. The team must contain at least 4 chefs who are skilled in preparing meat:
$$x_1 + x_2 + x_3 + x_{12} + x_{15} + x_{16} + x_{17} + x_{18} \geq 4.$$
4. The team must contain at least 3 chefs who are skilled in preparing dessert.
$$x_3 + x_4 + x_8 + x_{10} + x_{20} \geq 3.$$
5. To promote young people, the team must contain at least 2 chefs who are 20 or less.
$$x_8 + x_9 + x_{14} + x_{18} \geq 2.$$
6. Yoda and Obi-Wan cannot stand each other, and it is not possible to have both of them in the team.
$$x_{12} + x_{19} \leq 1.$$
7. Han and Sheev have so much experience in working together that if one of the two is included in the team, the other must be too.
$$x_5 = x_{16}.$$
8. For the sake of fairness, not more than 3 chefs from the same restaurant should be hired in the same team. We observe that 4 candidates come from Crissier but this is the only restaurant with more than 3 chefs.
$$x_3 + x_5 + x_6 + x_{16} \leq 3.$$

Thus, the optimization problem is
$$\max_{x \in \{0,1\}^{20}} \sum_{i=1}^{20} c_i x_i$$

subject to

$$
\begin{aligned}
x_5 + x_9 + x_{11} + x_{19} &\geq 3 \\
x_1 + x_2 + x_6 + x_7 + x_9 + x_{12} + x_{14} + x_{19} &\geq 4 \\
x_1 + x_2 + x_3 + x_{12} + x_{15} + x_{16} + x_{17} + x_{18} &\geq 4 \\
x_3 + x_4 + x_8 + x_{10} + x_{20} &\geq 3 \\
x_8 + x_9 + x_{14} + x_{18} &\geq 2 \\
x_{12} + x_{19} &\leq 1 \\
x_5 &= x_{16} \\
x_3 + x_5 + x_6 + x_{16} &\leq 3 \\
x_i \in \{0, 1\} \quad \forall i &= 1, ..., 20.
\end{aligned}
$$